

CHAPTER 2

2 SURVEY OF TECHNOLOGY

1. Safety at Your Fingertips: The History of and Case for Panic Buttons

Title: Safety at Your Fingertips: The History of and Case for Panic Buttons

Author: CM3 Building Solutions — Editorial Team

Publication Year: 2026

Publisher/Source: cm3inc.com

Summary:

This reference explains how panic buttons have developed from simple physical alarms to modern digital safety solutions. A panic button provides a quick way for a person to indicate that an emergency has occurred.

This concept is useful for Raksha because the system provides an SOS feature through which a user can report an emergency incident. The incident can contain information such as its type, category, description, priority, status and available location details.

2. Cell Phone Personal Safety Alarm

Title: Cell Phone Personal Safety Alarm

Author: Erik N. Hagenstad

Publication Year: 2016

Publisher/Source: United States Patent and Trademark Office (USPTO), U.S. Patent No. 9,451,061

Summary:

This reference describes a mobile safety system that combines different emergency features such as a panic button, warning functions, communication and location support.

The concept is related to Raksha because the project also tries to bring important emergency functions into one system. In Raksha, a user can create an emergency incident and the system can store the related incident and location information for further processing.

3. Multifunctional Personal Emergency Safety Device

Title: Multifunctional Personal Emergency Safety Device

Author: Jason L. Lassiter

Publication Year: 2017

Publisher/Source: United States Patent and Trademark Office (USPTO), U.S. Patent No. 9,697,720

Summary:

This reference describes an emergency device where one emergency action can start several safety-related functions.

The idea is useful for Raksha because an emergency report can involve different types of information. The system maintains incident details, user information, location, emergency contacts, notifications and response-related records in an organized way.

4. Global Positioning System (GPS)

Title: Global Positioning System (GPS)

Author: Multiple contributors

Publication Year: 2026

Publisher/Source: Wikipedia, The Free Encyclopedia

Summary:

GPS and other location technologies are used to determine the geographical position of a device. Location information is especially useful in emergency situations because it helps identify where an incident has been reported.

Raksha uses the available location information from the user's device through the Browser/Device Geolocation API. The system can store latitude and longitude along with other available location details. The accuracy of the location depends on device capability, permissions, network conditions and available location services.

5. Brief History of GPS

Title: Brief History of GPS

Author: The Aerospace Corporation

Publication Year: Not specified in the supplied reference

Publisher/Source: aerospace.org

Summary:

This reference explains the development of satellite-based navigation and location technology. Location services have become a common part of modern web and mobile applications.

For Raksha, location information is useful for connecting an emergency incident with a geographical position. The available latitude and longitude can be stored and later displayed or used by the administrator while managing incidents.

6. Who Invented GPS? Powerful Minds Behind the History of GPS

Title: Who Invented GPS? Powerful Minds Behind the History of GPS

Author: History Cooperative — Editorial Team

Publication Year: 2023

Publisher/Source: historycooperative.org

Summary:

This reference provides a historical overview of the researchers and developments that contributed to GPS technology.

The study of GPS is relevant to Raksha because location is an important part of emergency management. When a user reports an incident, the system can associate the available location information with that incident and make it easier for the administrator to understand the reported location.

7. WhatsApp

Title: WhatsApp

Author: Jan Koum and Brian Acton — founders; source is Wikipedia

Publication Year: 2026

Publisher/Source: Wikipedia, The Free Encyclopedia

Summary:

WhatsApp is a commonly used communication platform that allows users to exchange messages and other information.

WhatsApp was studied because messaging applications can be useful for emergency communication. In Raksha, WhatsApp-based communication can be supported through a configured provider where available. It is treated as a communication option rather than assuming that an external WhatsApp service will always be available.

8. The History of WhatsApp Business API

Title: The History of WhatsApp Business API

Author: Qiscus — Editorial Team

Publication Year: 2026

Publisher/Source: qiscus.com

Summary:

The WhatsApp Business API provides a way for organizations and software systems to communicate with users through automated messages.

This technology was studied because an emergency management system may require communication with users or contacts. In Raksha, WhatsApp communication can be connected through a configured communication provider if the required service is available and properly configured.

9. Twilio

Title: Twilio

Author: Jeff Lawson, Evan Cooke and John Wolthuis

Publication Year: 2026

Publisher/Source: Wikipedia, The Free Encyclopedia

Summary:

Twilio is a cloud communication platform that provides programmable communication services such as SMS and voice.

This technology was surveyed because emergency systems may require external communication services. Raksha supports the concept of configurable communication providers for services such as email, SMS or WhatsApp, depending on the services configured for the project.

10. How Jeff Lawson Founded Twilio: Build with Conviction or Risk Burnout

Title: How Jeff Lawson Founded Twilio: Build with Conviction or Risk Burnout

Author: Kyle Kelly-Yahner

Publication Year: 2015

Publisher/Source: Twilio Inc. — Company Blog

Summary:

This reference discusses the development of Twilio and the use of APIs to provide communication services to software applications.

The main idea is useful for Raksha because APIs can connect a web application with external services. Raksha uses a REST API-based backend, while external communication providers can be configured when required.

11. Evolution of Speech Recognition Technology — From Audrey to Siri

Title: Evolution of Speech Recognition Technology — From Audrey to Siri

Author: Verbit — Editorial Team

Publication Year: 2019

Publisher/Source: verbit.ai

Summary:

This reference explains how speech recognition developed from early systems into modern voice-based applications.

Speech recognition was studied because voice-based emergency interaction can be useful in situations where a person may not be able to operate a device normally. It can be considered as an additional feature for future versions of Raksha.

12. Nervous System: Audrey and the Dragon — The History of Voice Recognition Technology

Title: Nervous System: Audrey and the Dragon — The History of Voice Recognition Technology

Author: David Kalat

Publication Year: 2021

Publisher/Source: Berkeley Research Group (BRG) Insights

Summary:

This reference describes the early development of speech recognition and how computers gradually became better at understanding spoken words.

The study is relevant to Raksha because voice-based emergency interaction can provide an alternative method of communication. However, it is considered an optional enhancement and is not treated as a required part of the core web application

13. Google Translate

Title: Google Translate

Author: Franz Josef Och — associated with the original Google Translate architecture

Publication Year: 2026

Publisher/Source: Wikipedia, The Free Encyclopedia

Summary:

Translation technology allows information to be converted from one language into another. This can be useful in emergency situations where people may communicate in different languages.

The technology was studied for Raksha because multilingual support can make a safety system easier to use. Automated translation through an external service can be added as an enhancement when the required service is configured.

14. Celebrating 20 Years of Google Translate: Fun Facts, Tips and New Features to Try

Title: Celebrating 20 Years of Google Translate: Fun Facts, Tips and New Features to Try

Author: Rose Yao

Publication Year: 2026

Publisher/Source: Google Blog / The Keyword

Summary:

This reference explains the development of Google Translate and its support for different languages.

Multilingual communication can be useful in disaster management because users may prefer to communicate in their local language. Therefore, translation technology was studied as a possible future enhancement for Raksha.

15. Decision Tree and Random Forest Algorithms: Decision Drivers

Title: Decision Tree and Random Forest Algorithms: Decision Drivers

Author: History of Data Science — Editorial Team

Publication Year: 2022

Publisher/Source: historyofdatascience.com

Summary:

This reference explains Decision Tree and Random Forest algorithms and their use in classification problems.

Machine learning can be useful in emergency applications for analysing information and identifying possible incorrect or accidental alerts. The concept was studied for Raksha as a possible way to improve future intelligent alert analysis.

16. Random Forests

Title: Random Forests

Author: Leo Breiman

Publication Year: 2001

Publisher/Source: Machine Learning Journal, Springer

Summary:

Leo Breiman's research introduced the Random Forest method, which combines multiple decision trees for classification.

This research provides a useful academic background for understanding machine-learning-based classification. For Raksha, such techniques may be used in future development for analysing emergency data or reducing false alerts, depending on the final implementation.

17. MongoDB and Mongoose

Technology: MongoDB

Database Type: NoSQL Document Database

ODM: Mongoose

Summary:

MongoDB is a document-based database used for storing application data. Mongoose works with MongoDB in Node.js applications and helps define schemas, validate data and manage database operations.

MongoDB is suitable for Raksha because the system contains different types of information such as users, emergency contacts, incidents, locations, notifications, rescue teams, rescue assignments, facilities, risk information and reports.

The Raksha database is organized into separate collections. References between related documents are maintained at the application level using Mongoose. This makes the database suitable for managing the different modules of the system.

18. Node.js and Express.js

Technology: Node.js

Framework: Express.js

API Architecture: REST API

Summary:

Node.js provides the runtime environment for the backend of a web application. Express.js can be used with Node.js to create APIs and handle requests from the frontend.

Raksha uses Node.js and Express.js as its backend technologies. The backend manages application requests, communicates with MongoDB, handles user authentication and supports different system modules through REST APIs.

This technology is important because the frontend and database need a backend layer through which data can be processed securely and consistently. The original survey also identified Node.js and Express.js as suitable backend technologies for Raksha.

19. React.js, TypeScript, Vite and Tailwind CSS

Frontend: React.js

Programming Language: TypeScript

Build Tool: Vite

Styling: Tailwind CSS

Summary:

React.js is used for developing interactive web interfaces using reusable components. TypeScript adds type checking and helps make the frontend code easier to maintain. Vite provides a fast development and build environment, while Tailwind CSS is used for designing the user interface.

These technologies are suitable for Raksha because the system contains multiple screens for users and administrators. These include login, user profile, SOS/incident management, emergency contacts, emergency resources, notifications, risk information, admin dashboard and reports.

The use of React.js makes Raksha a web-based application, rather than a React Native mobile application. This is consistent with the software requirements of the project.

20. Python, FastAPI and AI Service Integration

AI Backend: Python

AI Framework: FastAPI

AI Integration: Configured AI Service API

Summary:

Python is widely used for developing data-processing and artificial-intelligence-related applications. FastAPI provides a framework for creating APIs using Python.

These technologies are relevant to Raksha because the project includes an AI-related backend/service layer in its software requirements. A configured AI service can be connected to the main backend through an API when required.

The AI component is kept separate from the main Node.js backend so that AI-related processing can be managed independently. This structure also makes it easier to modify or extend the AI functionality in the future.

21. OpenStreetMap, Leaflet and Location-Based Services

Mapping: OpenStreetMap

Map Library: Leaflet.js

Location: Browser/Device Geolocation API

Map Service: Configured Map Service

Summary:

OpenStreetMap provides geographical map data, while Leaflet.js is a lightweight JavaScript library used to display interactive maps.

These technologies are useful for Raksha because location is an important part of emergency and disaster management. The system can use available device location information and display relevant incident or risk-related locations on the administrative interface.

Leaflet can also help display location-based information such as incident locations and risk zones in a simple visual form. OpenStreetMap can be used as the map-data source where

configured. The original technology survey also identified OpenStreetMap and Leaflet as useful technologies for the Raksha dashboard.

Summary: Raksha is a web-based Women Safety and Disaster Management System. It uses React.js, TypeScript, Vite and Tailwind CSS for the frontend, Node.js and Express.js for the backend, and MongoDB with Mongoose for data storage. The system uses location services and map technologies for emergency location management. JWT and bcrypt support authentication and password security. Python and FastAPI are used for the AI-related service layer, while communication and other advanced technologies can be configured when required. The technology survey helped in selecting suitable technologies for developing and managing the Raksha system.